

Force and gravity

Answers

Weight

- The weight of an object is the force acting on the object in a gravitational field. The stronger the gravitational field, the greater the weight of a body. Weight is measured in the units of newtons (N)

The weight (W) of a body of mass (m) in a gravitational field (g = acceleration due to gravity) is given by the formula:

$$W = mg$$

On the Earth's surface ' g ' typically has a value of 9.8 m/s^2 but this varies with altitude and latitude. On other planets, the value of ' g ' also changes.

Complete the table using the above formula.

Mass (kg)	Gravity (m/s^2)	Weight (N)
20	9.8	$20 \times 9.8 = 196$
30	$\frac{286.5}{30} = 9.55$	286.5
$\frac{400}{9.8} = 40.8$	9.8	400.0

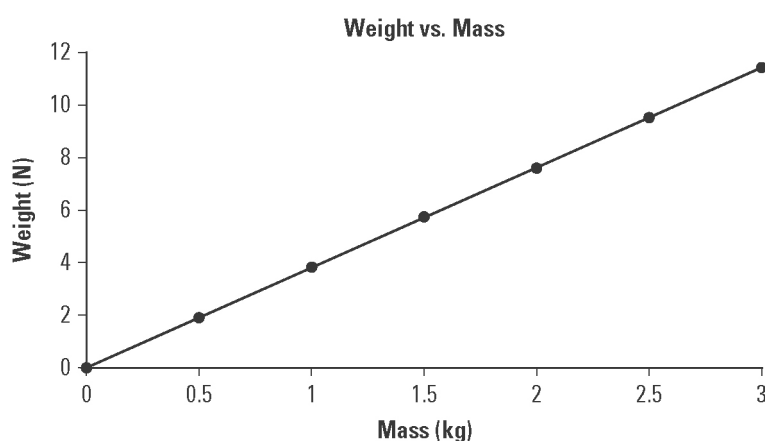
Spreadsheets and graphing

- A spreadsheet program such as Excel is a useful computer application for tabulating data and plotting graphs. In this exercise you will practise using a spreadsheet application.

The following mass and weight data was collected from the planet Mars. The data is presented as you would set it out in the spreadsheet application.

- Open the spreadsheet application and enter the data as tabulated below.
- Select *Insert* from the menu bar.
- Select *Scatter* as the chart type. Select 'Scatter with straight lines and markers'.
- The *Design* and *Layout* tabs can be used to insert a chart title and then each axis title and unit.
- Select the gridlines you would like to show.

	A	B
1	Mass (kg)	Weight (N)
2	0.0	0.0
3	0.5	1.9
4	1.0	3.8
5	1.5	5.7
6	2.0	7.6
7	2.5	9.5
8	3.0	11.4



Use your chart to answer the following questions.

- (f) Estimate the weight of a 1.3 kg body on the surface of Mars.4.9 N.....
- (g) Estimate the mass of an object that weighs 10.0 N on Mars.2.6 kg.....
- (h) Suggest what information the slope of the graph may provide.

(Clue: The formula $W = mg$ can be rearranged as $g = \frac{W}{m}$.)

The slope provides the acceleration due to gravity on the planet.....
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